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Original Article

Outcomes of Orbital Exenteration for Craniofacial Lesions

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BACKGROUND: Orbital exenteration (OE) is an ablative procedure used in the management of malignancies of the orbit of either primary or secondary origin. Publications evaluating this procedure have suffered from small patient numbers, heterogeneity of pathologies, and poor patient follow-up. The purpose of this study was to assess patient outcomes in a large cohort of patients undergoing OE at a tertiary cancer center. **METHODS:** A retrospective review was conducted of 180 consecutive patients who underwent OE at the authors' institution. Overall survival (OS) was the primary end point measured in the study. Time to locoregional recurrence (progression-free survival [PFS]) and disease-free survival were secondary end points. **RESULTS:** Between the years 1993 and 2011, 180 consecutive patients received OE for craniofacial malignancy at the authors' institution. The median follow-up for the cohort was 9.7 years (116 months). The median OS was 73 months, and the median PFS was 96 months. The presence of perineural invasion was associated with shorter OS ($P = .01$) and PFS ($P < .01$). Magnetic resonance imaging was predictive of perineural invasion ($P < .01$). Positive margins were associated with shorter PFS than negative margins ($P < .01$) but with no change in OS ($P = .15$). The overall complication rate was 15%. The major complication rate (Clavien-Dindo 3b or greater) was 2.8% (n = 5), and there was 1 death observed (0.6%). **CONCLUSIONS:** Used judiciously in the setting of a multidisciplinary management plan, OE for tumor control is a safe therapy. ***Cancer* 2021;127:2465-2475**.
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LAY SUMMARY:

- Between the years 1993 and 2011, 180 consecutive patients received orbital exenteration for craniofacial malignancy at the MD Anderson Cancer Center. The median follow-up for the cohort was 9.7 years.
- The presence of perineural invasion was associated with shorter overall survival ($P = .01$) and progression-free survival ($P < .01$). Magnetic resonance imaging was predictive of perineural invasion ($P < .01$).
- Positive margins were associated with shorter progression-free survival than negative margins ($P < .01$). The overall complication rate was 15%. The major complication rate (Clavien-Dindo 3b or greater) was 2.8% (n = 5).

KEYWORDS: craniofacial malignancy, locoregional recurrence, orbital exenteration, orbital tumors, overall survival.

INTRODUCTION

Orbital exenteration (OE) is the surgical excision of all the soft tissues of the orbit, including the globe, extraocular muscles, lacrimal gland, and optic nerve.[1] Its primary indication is for resection of malignant neoplasms of primary or secondary origin that involve the orbit. This procedure results in monocular vision loss and significant cosmetic alterations to the face.[2] At our institution, a multidisciplinary surgical team, consisting of ophthalmology, head and neck surgery, plastic and reconstructive surgery, and neurosurgery, helps in selecting patients and performing these procedures. Historically, involvement of the muscle cone by malignancy almost always necessitates OE if curative surgery is to be attempted. Early experience with craniofacial resections for sinonasal malignancy demonstrated better locoregional control, lower recurrence rates, and better survival rates when the orbit was removed.[3,4] In more recent years, multidisciplinary eye-sparing approaches using a combination of eye-sparing surgery and subsequent adjuvant radiation therapy or concurrent chemoradiation in the case of lacrimal gland carcinoma and biologic therapy and immunotherapy in the case of cutaneous cancers have also demonstrated efficacy. Furthermore, several reports have shown that OE may not be necessary to improve survival for early-stage disease.[5-11] In addition, neoadjuvant chemotherapy for sinonasal cancers may be used for cytoreduction, and in patients with a significant response to chemotherapy, orbital preservation is feasible without compromising the oncologic outcome for select patients.[12,13] Despite these advances in eye-preserving treatments, OE is still necessary in a subset of patients with locally advanced cancers

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with extension into the orbit or with locally advanced (T4) primary cancers of the orbit. Herein, we present a retrospective review of the outcomes and operative techniques for 180 patients undergoing OE at a large tertiary cancer center and discuss the variables contributing to disease recurrence and survival.

MATERIALS AND METHODS

Design, Setting, and Participants

We identified and reviewed the medical records of 180 consecutive patients who underwent OE at our institution between 1993 and 2011 with a review period through 2020. This was performed under an institutional review board–approved protocol in compliance with institutional regulations with regard to the study of human subjects. Consent was obtained for all described procedures, and waivers of informed consent and authorization were granted for the use of secondary research data. The patients were identified through a search of the prospectively collected brain and spine tumor database as well as the head and neck and ophthalmology departmental databases at our institution. Patient demographics, preoperative imaging, operative reports, histopathologic reports, and perioperative management documentation were procured and reviewed. Patients undergoing orbital-sparing surgeries were excluded. Selection for OE consists of a multistep process and has evolved over time on the basis of the effectiveness of available adjuvant therapies. In general, if the surgical team feels that a total resection can be technically achieved, then surgery is recommended. If the disease is extraconal, then a direct surgical evaluation of the orbit is undertaken to determine the depth of invasion. If there is neoplastic involvement of the muscle cone or unappreciated intraconal invasion, then orbitectomy is performed. Orbital apex involvement almost always necessitates OE. Reflecting the literature on the subject, our institution has increasingly used orbital-preservation techniques for orbital tumors with limited extraconal disease. Part of the purpose of this study was to set a benchmark expectation for oncologic outcomes to allow comparison with our expanding cohort of patients undergoing orbital preservation and aggressive multimodal therapy.

Operative Technique

The operative technique for each patient was individualized on the basis of pathology and location; however, it generally fell into 1 of 4 categories: OE alone (with or without eyelid sparing), facial-orbital exenteration, cranio-orbital exenteration, or craniofacial OE (Figs. 1 and 2). Additional information on the operative technique is provided in the supporting information.

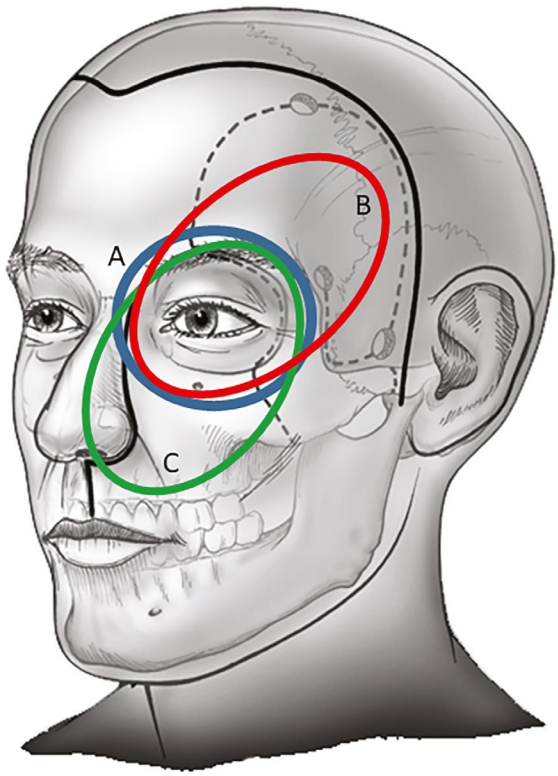


Figure 1. Artist's representation of the 3 main approaches for orbital exenteration: (A) transorbital, (B) cranio-orbital, and (C) facial-orbital. Property of the Department of Neurosurgery, MD Anderson Cancer Center, used with permission.

Statistical Analysis

Overall survival (OS) was the primary end point measured in this study. Progression-free survival (PFS) and disease-free survival (DFS) were secondary end points measured. Details on the statistical analysis are provided in the supporting information.

RESULTS

Patient and Tumor Characteristics

Patient characteristics are summarized in Table 1. On the basis of preoperative imaging, which was available for 172 patients, the tumor was only intraconal in 8 patients (5%), only extraconal in 118 patients (69%), and both intraconal and extraconal in 39 patients (23%). The orbital apex was involved in 62 patients (36%). One hundred of 177 total patients with available data (56%)

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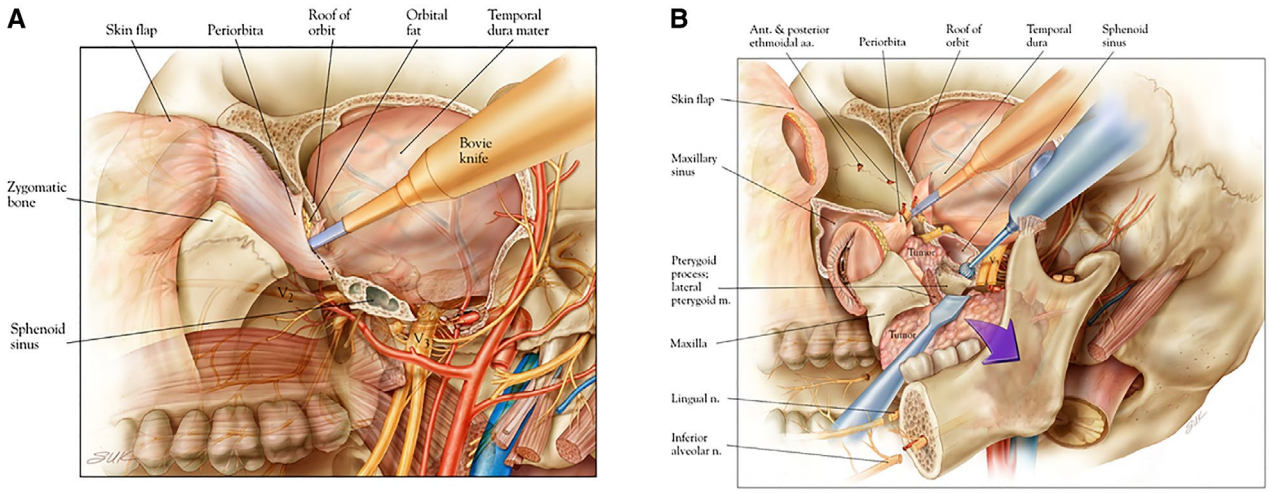


Figure 2. Artist's representation of cranio-orbital resection. (A) The orbital apex is divided just anterior to the superior orbital fissure. (B) A combined resection of the orbital, maxilla, and infratemporal fossa for an extensive maxillary sinus tumor. Property of the Department of Neurosurgery, MD Anderson Cancer Center, used with permission.

TABLE 1. Patient Characteristics

Characteristic		Value
Total, No. (%)		180 (100)
Sex, No. (%)	Male	113 (63)
	Female	67 (37)
Age, median (range), y		63 (6-94)
Side of exenteration, No. (%)	Right	90 (50)
	Left	90 (50)
KPS, median (range)		90 (70-100)
Preoperative pain score, No. (%)[a]	Mild (0-3)	64 (67)
	Moderate (4-6)	19 (20)
	Severe (7-10)	13 (13)
Preoperative vision, No. (%)[b]	Normal	35 (48)
	Mild deficit (20/60-20/100)	6 (8)
	Moderate deficit (20/100-20/400)	0 (0)
	Severe deficit (hand motion, light perception)	18 (25)
	Blind	14 (19)
Preoperative proptosis, No. (%)		47 (27)
Preoperative cranial neuropathy, No. (%)		136 (76)
Preoperative radiation, No. (%)		39 (22)
Preoperative chemotherapy, No. (%)		36 (20)
Postoperative radiation, No. (%)		107 (59)
Postoperative chemotherapy, No. (%)		34 (18)

Abbreviations: KPS, Karnofsky performance scale.
[a]Data were available for 96 patients.
[b]Data were available for 73 patients.

had invasion into the bone surrounding the orbit. There was extracranial extension as follows: infratemporal fossa (n = 41), nasal cavity (n = 51), ethmoid sinus (n = 52), maxillary sinus (n = 50), frontal sinus (n = 35), and sphenoid sinus (n = 18). Additionally, there was epidural spread in 49 patients (28%), cavernous sinus extension in 24 patients (14%), and brain invasion in 6 patients (3%; Table 2). Perineural spread was identified on preoperative magnetic resonance imaging in 27 of 159 patients (17%). Twenty-one of these 27 patients (78%) had no known symptoms consistent with these perineural findings (Fig. 3). Additionally, 8 pediatric patients (age < 18 years at the time of surgery) were treated for a variety of histologies, including sarcoma (n = 3), retinoblastoma (n = 1), adenoid cystic carcinoma (n = 1), neurofibroma (n = 1), metastatic melanoma (n = 1), and carcinoma ex pleomorphic adenoma (n = 1). No deaths were observed in these patients. Recurrence was observed in only 1 patient in the form of distant metastases for the patient with a diagnosis of carcinoma ex pleomorphic adenoma.

TABLE 2. Tumor Extensions

Location	No. (%)[a]	Associated With Positive Margins
Intraconal	47 (27)	No
Extraconal	157 (91)	No
Orbital apex	62 (35)	$P < .001$
Infratemporal fossa	41 (23)	$P < .012$
Nasal cavity	51 (30)	No
Ethmoid sinus	52 (31)	No
Maxillary sinus	50 (30)	No
Frontal sinus	35 (21)	No
Sphenoid sinus	18 (11)	No
Epidural	49 (28)	$P < .001$
Cavernous sinus	24 (14)	$P < .001$
Brain parenchyma	6 (3)	$P < .009$

[a]When data were available.

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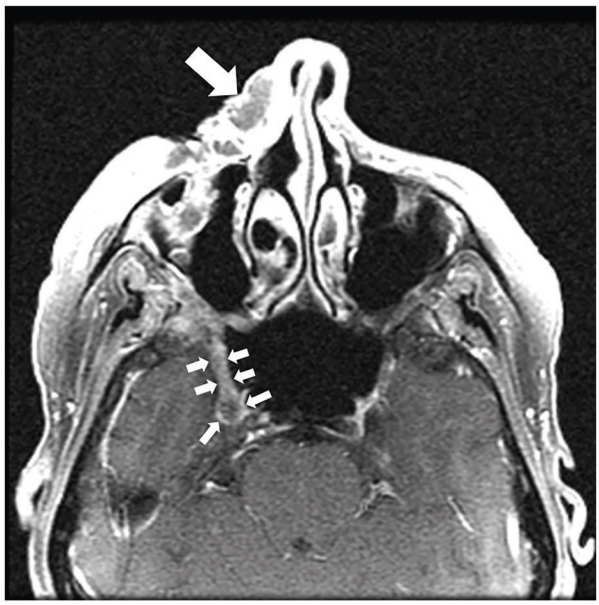


Figure 3. Axial postcontrast T1-weighted magnetic resonance imaging identifies an invasive skin cancer extending into the inferior orbit (large arrow). There is evidence of perineural tumor extension along the second division of the trigeminal nerve (small arrows).

TABLE 3. Histology

Histology		No. (%)
Squamous and basosquamous cell carcinoma	Periorbital skin	41 (23)
	Sinonasal	13 (7)
	Primary orbital	8 (4)
	Distant metastases	1 (0.6)
Adenoid cystic carcinoma	Lacrimal gland	17 (9)
	Other	7 (4)
Sarcoma	Leiomyosarcoma	1 (<1)
	Chondrosarcoma	2 (1)
	Rhabdomyosarcoma	2 (1)
	Synovial sarcoma	1 (<1)
	Alveolar soft part sarcoma	1 (<1)
	Chondroblastic osteosarcoma	1 (<1)
	Osteoblastic osteosarcoma	4 (2)
	Liposarcoma	1 (<1)
	Undifferentiated pleomorphic sarcoma	3 (2)
	Sarcoma, NOS	5 (3)
Melanoma	Non-ocular	17 (9)
	Ocular	3 (2)
Basal cell carcinoma (skin)		18 (10)
Mucoepidermoid carcinoma		2 (1)
Sebaceous carcinoma		7 (4)
Carcinoma, NOS		4 (2)
Adenocarcinoma		7 (4)
Neuroendocrine carcinoma		3 (2)
Meningioma		3 (2)
Osteoradionecrosis		3 (2)
Ameloblastoma		2 (1)
Retinoblastoma		1 (<1)
Neurofibroma		1 (<1)

Abbreviation: NOS, not otherwise specified.

TABLE 4. Histology of Primary Orbital Tumors

Histology	No. (%)
Total	46 (100)
Adenoid cystic carcinoma	17 (37)
Squamous/basosquamous cell carcinoma	8 (17)
Adenocarcinoma	5 (11)
Non-ocular melanoma (conjunctival/orbital)	5 (11)
Ocular melanoma (uveal)	3 (7)
Sarcoma	3 (7)
Carcinoma, NOS	3 (7)
Retinoblastoma	1 (7)
Treatment effect	1 (7)

Abbreviation: NOS, not otherwise specified.

OE was performed for diverse pathologies, including squamous cell or basosquamous carcinoma (SCC; n = 63), adenoid cystic carcinoma (n = 24), sarcoma (n = 22), melanoma (n = 20; 17 non-ocular cases and 3 ocular cases), basal cell carcinoma (BCC; n = 18), carcinoma otherwise unspecified (n = 13), adenocarcinoma (n = 7), meningioma (n = 3), neuroendocrine carcinoma (n = 3), treatment effect (n = 3), ameloblastoma (n = 2), retinoblastoma (n = 1), and neurofibroma (n = 1; Table 3).

Primary orbital tumors occurred in 46 of 180 total patients (26%) in the cohort (Table 4). Tumors that originated outside the orbit but invaded the orbit or were metastatic to the orbit were designated as secondary orbital tumors, which occurred in 134 patients (74%; Table 5). Of the 134 secondary orbital tumors, 76 (57%) were an extension of a skin malignancy, 40 (30%) were an extension of a sinonasal malignancy, 9 (7%) were an extension of a bone malignancy of the skull base, 7 (5%) were an extension of an intracranial malignancy, and 2 (1%) were metastatic to the orbit (SCC and non-ocular melanoma; Table 6).

Four approaches were used: direct transorbital OE, facial-orbital exenteration, cranio-orbital exenteration, and a combination of these approaches (Fig. 1). OE alone was performed in 63 of 180 total patients (35%). Facial-orbital exenteration was performed in 26 of the 180 patients (14%), cranio-orbital exenteration was performed in 31 (17%), and combined craniofacial OE was performed in 59 patients (33%).

The approach to flap coverage of OE defects at our institution has been described.[14] A microvascular free flap was used in 127 of the 180 surgeries (71%), a split-thickness skin graft was used in 42 surgeries (23%), a temporalis rotational flap was used in 5 patients (3%), and eyelid preservation was performed in 10 surgeries (6%). No difference in medical complications ($P = .85$) or surgical complications ($P = .28$) at 30 days postoperatively was identified in patients undergoing reconstruction with local flaps in comparison with microvascular

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TABLE 5. Histology of Secondary Orbital Tumors

Histology		No. (%)
Sinonasal malignancy	Total	40 (22)
	Squamous/basosquamous cell carcinoma	13 (7)
	Adenocystic carcinoma	7 (4)
	Sarcoma	7 (4)
	Carcinoma, NOS	3 (2)
	Adenocarcinoma	2 (1)
	Ameloblastoma	2 (1)
	Neuroendocrine carcinoma	2 (1)
	Non-ocular melanoma	2 (1)
	Basal cell carcinoma	1 (<1)
	Treatment effect	1 (<1)
Skin malignancy	Total	76 (42)
	Squamous/basosquamous cell carcinoma	41 (23)
	Basal cell carcinoma	17 (9)
	Non-ocular melanoma	9 (5)
	Carcinoma, NOS	7 (4)
	Sarcoma	1 (<1)
	Treatment effect	1 (<1)
	Sarcoma	2 (1)
	Neuroendocrine tumor	1 (<1)
	Neurofibroma	1 (<1)
Metastatic malignancy	Total	2 (1)
	Squamous/basosquamous cell cancer	1 (<1)
	Non-ocular melanoma	1 (<1)
Nonsinonasal head and neck malignancy	Total	9 (5)
	Sarcoma	9 (5)
	Intracranial malignancy	7 (4)
	Meningioma	3 (2)

Abbreviation: NOS, not otherwise specified.

TABLE 6. Primary Versus Secondary Orbital Tumors

Primary vs Secondary		No. (%)
Primary orbital tumors		46 (26)
Secondary orbital tumors	Total	134 (74)
	Extension of skin malignancy	76 (42)
	Extension of sinonasal malignancy	40 (22)
	Extension of cranial/skull base malignancy	9 (5)
	Extension of intracranial malignancy	7 (4)
	Metastatic to the orbit	2 (1)

reconstructions. During the course of the review period included in this study (1993-2011) we noted an increasing number of microvascular free flaps and fewer local tissue flaps.

Outcomes

For patients alive at the last follow-up, the mean and median length of follow-up was 9.7 years or 116 months (range, 0-296 months). For the entire study population, the median OS was 73 months (95% CI, 59-93 months). The median PFS was 96 months (95% CI, undefined), and the median DFS was 53 months (95% CI, undefined; Fig. 4). Specific OS, PFS, and DFS were assessed by tumor origin. Patients with primary orbital tumors had a median OS of 93 months (95% CI, undefined), whereas patients with a secondary orbital tumor had a median OS of 68 months (95% CI, 54-94 months; $P = .06$; Fig. 5). Patients with a primary orbital tumor had a median PFS of 96 months (95% CI, undefined), whereas patients with a secondary orbital tumor had a median PFS of 74 months (95% CI, undefined; $P = .56$). There was no significant difference in median DFS between patients with primary and secondary orbital tumors (53 months; 95% CI, undefined; $P = .96$).

Histology

The histology category (SCC, adenoid cystic carcinoma, non-ocular melanoma, BCC, or sarcoma) was a significant predictor of OS ($P < .01$) in the cohort but not of PFS ($P = .28$) or DFS ($P = .13$; Fig. 6). The SCC group had a median OS of 57 months (95% CI, 35-91 months), a median PFS of 74 months (95% CI, undefined), and a median DFS of 53 months (95% CI, undefined). The adenoid cystic carcinoma group had a median OS of 105 months (95% CI, undefined), a median PFS of 53 months (95% CI, undefined), and a median DFS of 41 months (95% CI, undefined). The non-ocular melanoma group had a median OS of 42 months (95% CI, undefined), a median PFS of 40 months (95% CI, undefined), and a median DFS of 9 months (95% CI, undefined). The median OS for patients with sarcoma was 186 months (95% CI, undefined). The median OS for patients with BCC was 103 months (95% CI, undefined). Median PFS and DFS were not reached for patients with BCC or sarcoma. A breakdown of outcomes in patients with sinonasal and skin epithelial tumors is summarized in Supporting Table S1.

Extent of Resection

A gross total tumor excision was accomplished in 172 of 180 total patients (96%). Near total resection was achieved in 8 patients (4%). Pathologic margins were microscopically positive in 82 patients (46%) and negative in 98 patients (54%). The median OS was 59 months (95% CI, 31-91 months) for patients with positive margins and 87 months (95% CI, 66-121 months) for those with negative margins ($P = .15$). The median PFS for patients with positive margins was 49 months (95% CI, undefined), whereas the median PFS for those patients with negative margins was not reached ($P < .01$). The median DFS for patients with positive margins

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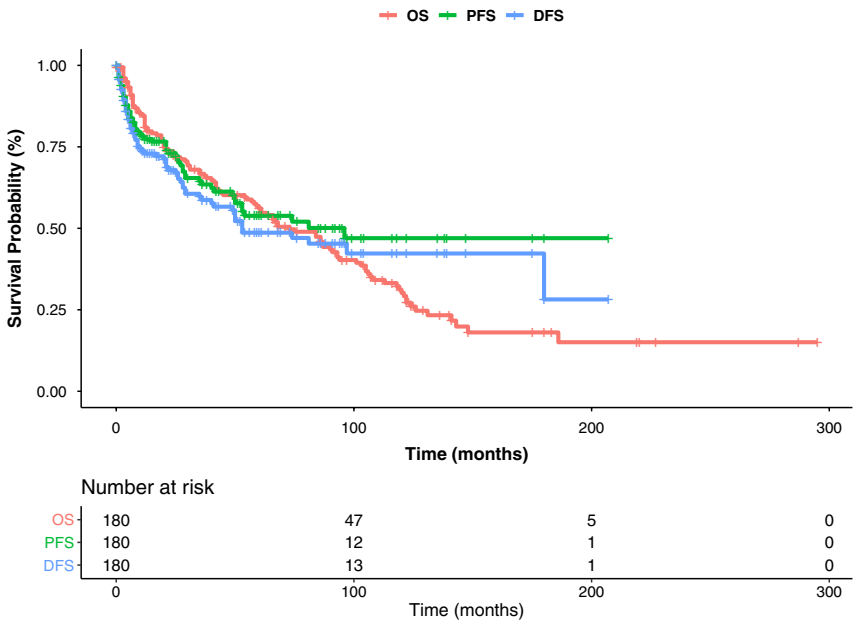


Figure 4. Kaplan-Meier curves representing OS, PFS (defined as the time to locoregional recurrence), and DFS (defined as the time to locoregional or distant recurrence). DSF indicates disease-free survival; OS, overall survival; PFS, progression-free survival.

was 29 months (95% CI, 21-81 months), whereas the median DFS for those patients with negative margins was 180 months (95% CI, undefined; $P < .01$; Fig. 7).

Perineural Invasion

The median OS was 57 months (95% CI, 29-93 months) for patients with pathologic evidence of perineural invasion and 87 months (95% CI, 66-121 months) for those without it ($P < .01$). The median PFS was 53 months for those patients with pathologic evidence of perineural invasion (95% CI, undefined) and undefined for those without such evidence ($P = .01$). The median DFS was 28 months (95% CI, 17-81 months) for patients with pathologic evidence of perineural invasion and 180 months (95% CI, undefined) for those without it ($P < .01$; Fig. 8). Among the 180 patients, the histologic categories in which pathologic evidence of perineural invasion was identified were neurofibroma (1 of 1; 100%), adenoid cystic carcinoma (18 of 24; 75%), meningioma (2 of 3; 67%), adenocarcinoma (4 of 7; 57%), ameloblastoma (1 of 2; 50%), SCC (29 of 63; 46%), carcinoma not otherwise specified (3 of 13; 23%), non-ocular melanoma (6 of 17; 35%), and BCC (10 of 18; 56%). Additional information on patient outcomes is provided in the supporting information.

Complications

Overall, 27 patients (15%) suffered complications during the 30-day postoperative period that were categorized according to the Clavien-Dindo classification (grades I-V).[15] Complications are described in detail in the supporting information and are summarized in Table 7. The overall complication rate in the literature was 27.6% (95% CI: 21.3% – 34.9%) using a random effects model (Supporting Figure S1).

Pain

We attempted to quantify the levels of pain experienced by patients from orbital tumors both preoperatively and postoperatively. The majority of patients for whom data were available (67%) had no or only mild pain before surgery. Ten of the 13 patients presenting with severe pain (pain score, 7-10 of 10) had postoperative data available. Nine of these patients improved to mild or no pain (pain score, 0-3), and 1 improved to moderate pain (pain score, 4-6). For 13 of the 19 patients presenting with moderate pain, postoperative data were available, with 11 patients reporting improvement to no/mild pain and 2 patients reporting continuation of moderate pain.

DISCUSSION

OE is an invasive and disfiguring procedure that has been practiced for more than 100 years with descriptions in the literature as early as 1898.[16-23] The majority of cases are performed for life-threatening malignancy

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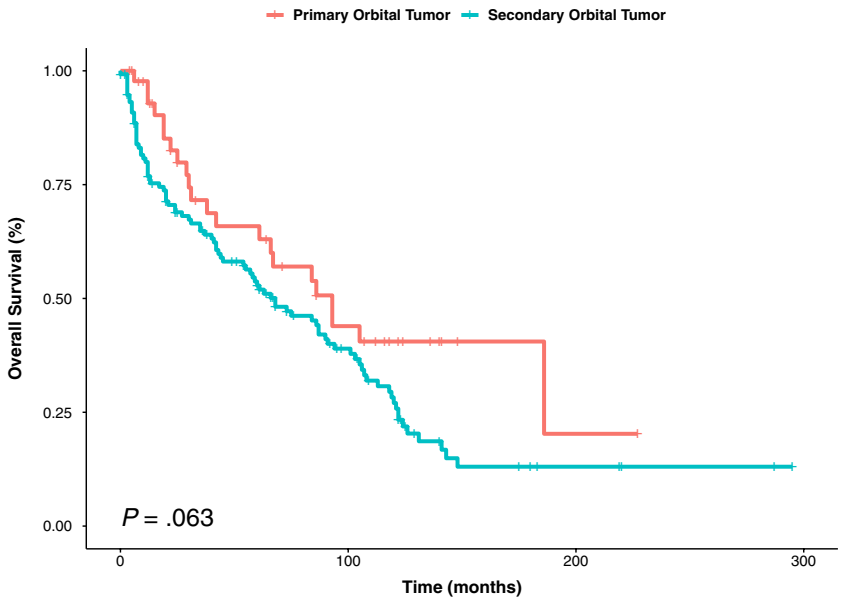


Figure 5. Kaplan-Meier representation of differences in overall survival between primary and secondary orbital tumors.

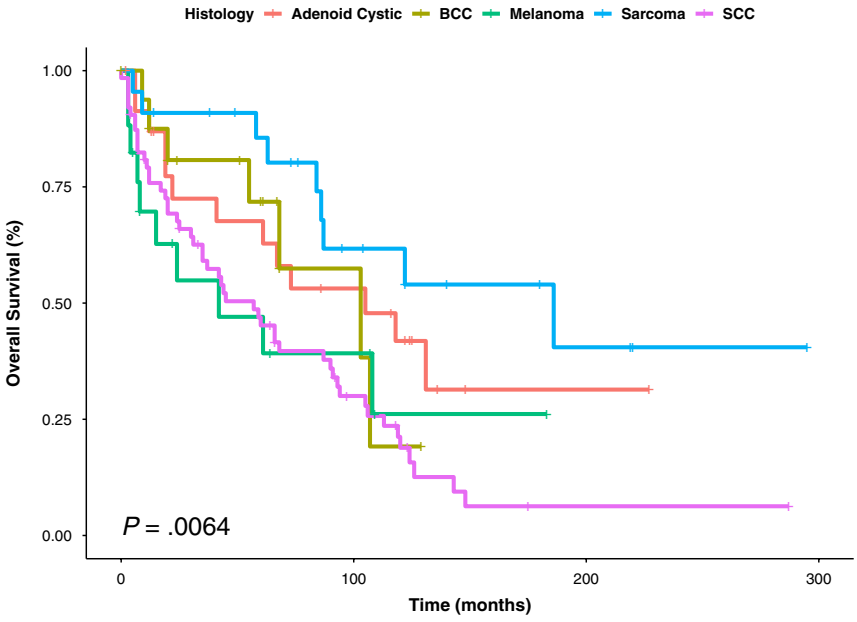


Figure 6. Kaplan-Meier representation of differences in overall survival between the 5 most common tumors with orbital invasion: adenoid cystic carcinoma, BCC, melanoma, sarcoma, and SCC. BCC indicates basal cell carcinoma; SCC, squamous cell/basosquamous cell carcinoma.

of the periorbital skin, paranasal sinuses, or skull base. With the introduction of orbital-preservation techniques, the indications for radical orbital excision have diminished, with increasing favor for preserving the involved eye over aggressive resection.[4-11,24-28] In 1994, Paridaens et al[29] argued that OE should be reserved as a palliative procedure for late disease stages. However, there has been concern about reduced locoregional

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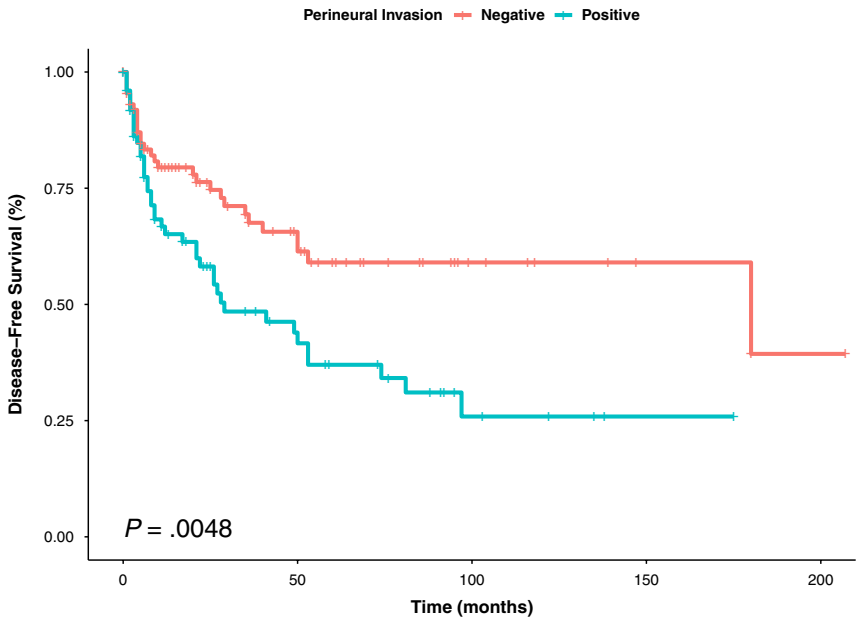


Figure 7. Kaplan-Meier representation of differences in disease-free survival between patients with positive and negative resected tumor sample margins on pathologic analysis.

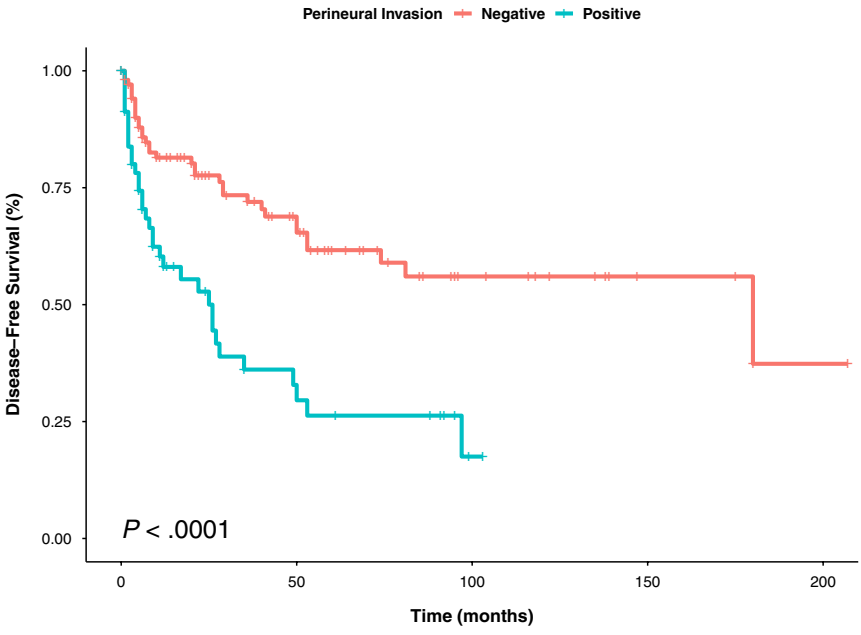


Figure 8. Kaplan-Meier representation of differences in disease-free survival between patients with and without perineural invasion observed on resected tumor samples.

control with preservation surgeries. A 2008 review of the literature by Suarez et al[24] concluded that orbital soft tissue–sparing techniques do not appear to adversely affect survival. Conversely, Hoffman et al[30] pointed out an important selection bias in studies comparing orbital sparing and OE in that more advanced tumors with worse prognoses are treated more aggressively with OE. They also argued that the deleterious effects of adjuvant

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TABLE 7. Complications by Postoperative Day 30

Complication	Descriptor	No. of Patients (%)
Medical	Total	8 (4)
	PNA	3 (2)
	PNA, PE, GI bleed	1 (<1)
	Arrythmia	1 (<1)
	Arrythmia, PNA	1 (<1)
	Seizure, PNA	1 (<1)
	Death	1 (<1)
Surgical	Total	19 (11)
	SSI, superficial	11 (6)
	SSI, deep	1 (<1)
	CSF leak	4 (2)
	Pneumocephalus	1 (<1)
	Cerebral edema	1 (<1)
	Epidural hematoma	1 (<1)
Clavien-Dindo classification[6]	Total	27 (15)
Grade I	NA	0 (0)
Grade II	PNA, PE, GI bleed, arrythmia, SSI	17 (9)
Grade IIIa	CSF leak	4 (2)
Grade IIIb	SSI requiring debridement	1 (<1)
Grade IVa	PNA with seizures requiring ICU care, EDH, pneumocephalus, cerebral edema	4 (2)
Grade IVb	NA	0 (0)
Grade V	Death (PNA + arrythmia)	1 (<1)
Age[a]	Total	27 (15)
	≥63 y	16
	<63 y	11
Surgical approach	Total	27 (15)
	Transorbital	9
	cranio-orbital	11
	facial-orbital	7
Reconstruction	Total	27 (15)
	Free flap	22
	Skin graft	4
	Eyelid-sparing	1
Previous treatment	Total	27 (15)
	Radiation only	2
	Chemotherapy only	3
	Chemoradiation	4

Abbreviations: CSF, cerebrospinal fluid; EDH, epidural hematoma; GI, gastrointestinal; ICU, intensive care unit; NA, not available; PE, pulmonary embolism; PNA, pneumonia; SSI, surgical site infection.
[a]The median age was 63 years for the cohort at the time of surgery.

TABLE 8. Previous Reports of Orbital Exenteration for Tumors

Source	No. of Patients	Complications, %
Bartley[49]	102	27
Levin 1991[38]	93	NA[a]
Rose 1994[39]	16	NA[a]
Paridaens 1994[29]	95	NA[a]
Mohr & Esser 1997[36]	77	26
Shields 2001[27]	56	21
Lin 2002[18]	7	NA[a]
Goldberg 2003[40]	25	28
Pushker 2004[21]	26	NA[a]
Rahman 2005[22]	69	35
Ben Simon 2005[23]	34	24
Nassab 2007[19]	32	53
Nemet 2007[20]	38	29
Kuo 2011[17]	38	42
Ackuaku-Dogbe 2011[37]	25	4
Zhang 2018[41]	102	NA[a]
Current series	180	15

Abbreviation: NA, not available.
[a]Complication rates were not mentioned.

radiotherapy following orbital-preservation techniques are not often acknowledged in these studies because it often results in either diminished or complete vision loss; however, there are data in the literature showing that the majority of well-selected patients, particularly those with less than T3 primary lacrimal gland carcinomas, are able to maintain their eye with better than 20/200 visual acuity, which is of great cosmetic and psychosocial importance to patients and their families.[7,8,10] With these advancements in mind, Hoffman et al concluded that OE should still have a role, particularly in cases of significant orbital bone invasion and in the absence of a significant response to neoadjuvant cytoreductive therapy. More recently, histology and the extent of local invasion have been included in management algorithms to better identify OE candidates.[31] In particular, these show a role for OE in patients with well-differentiated tumors with massive orbital content involvement or for high-grade tumors that are not responsive to biologic therapies or chemotherapies.[5,23,32] Similarly, combined craniofacial resection techniques are considered standard of care for invasive malignant tumors of the paranasal sinuses.[33-35]

Despite the general paradigm shift toward less aggressive resection techniques in the management of patients with head and neck tumors invading the orbit, patients undergoing OE survive for many years, even with malignant lesions. Retrospective studies of OE for cohorts of patients with varying tumor histologies and extents of invasion have reported 5-year OS rates ranging from 33% to 63%.[22,36]

To our knowledge, our study is the largest series of OE described to date, and it substantiates various factors that are indicative of outcome (previous series[17-23,27,29,36-41] are summarized in Table 8). Notably, our study confirmed that perineural invasion of the tumor had a statistically significant detrimental effect on PFS and OS in line with prior studies.[42] Retrograde perineural spread to the brainstem is a known behavior of many aggressive cutaneous malignancies, including melanoma and SCC,[43] and is often an indication for OE.[44] However, it is our experience that melanoma histology and the presence of perineural spread conferred worse outcomes not absolved by radical OE. Fortunately, the treatments for

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metastatic and/or locally advanced cutaneous melanomas have evolved significantly in the last decade with the advent of immunotherapy and its efficacy in many patients with metastatic and/or locally advanced orbital cutaneous melanoma who are responsive to immune checkpoint inhibitor therapy.[10] Additionally, the presence of positive margins was significantly associated with tumor extension into surrounding anatomical structures, including the infratemporal fossa, orbital apex, epidural space, cavernous sinus, orbital bone, and brain parenchyma. Positive margins were further predictive of local recurrence in our cohort with no effect on OS, and this indicated a potential benefit of radical resection over preservation techniques. Additionally, patients with negative margins had significantly longer DFS (180 vs 29 months; $P < .01$), and this indicated that margin-negative resection may in fact protect against distant metastatic disease. In a 2005 international collaborative study of 1193 patients treated with craniofacial resection of skull base tumors, Ganly et al[45] described an overall postoperative complication rate of 36.3%. Thus, our complication rates are comparable to, if not lower than, those accepted in the literature, and this suggests that OE can be performed safely in this patient population. However, this illuminates the importance of careful planning and patient selection when one is considering an appropriate therapy for patients with orbital malignancy. Additionally, it has been shown that elderly patients (≥70 years) often experience poorer overall outcomes than their younger counterparts for craniofacial resections of skull base malignancies,[46] although this was not seen to be a factor in our cohort of primarily older patients (median age 63 years).[47] The prolonged DFS observed in patients for whom margin-negative resection was achieved may indicate a protective effect against distant metastases, although this must be further evaluated in subsequent studies. Finally, the postoperative reduction in pain scores suggests that those patients with pain associated with their tumors will likely achieve improvement.

Social Challenges

After interviewing 12 patients and their family members following OE, Bonanno et al[2] found that acquaintances were sources of stigmatization and support alike as they intruded in the everyday existence of patients but also provided sympathy to cancer survivors. Facial disfigurement did not negatively affect interactions with family members and friends; however, spouses tended to display greater awareness of these changes and were more affected by the social difficulties experienced by patients in comparison with other family members. Despite these changes, Bonanno et al also found that the existence of cancer mitigated the negative effects of facial disfigurement. Furthermore, strategies to optimize appearance in the context of OE increasingly are being used with success.[36] A recent study by Mukoyama et al[48] found that early intervention by psychiatry decreased the psychological impact of the procedure and noted that Hospital Anxiety and Depression Scale scores normalized to the national average 2 years postoperatively.

Limitations

Our study is primarily limited by its retrospective design. As such, there was an evolution of the selection criteria and surgical approach over time. Although mitigated somewhat by our greater patient numbers and longer follow-up in comparison with other studies in the literature, the heterogeneity of the tumor histology, extent of invasion, and tumor location may reduce the generalization of our conclusions. Furthermore, there is an inherent bias that has been described for OE studies: patients with more advanced craniofacial malignancies with poorer perceived prognoses undergo more radical resections.[30] Despite these limitations, our study benefits from a large cohort size in comparison with other studies of its kind as well as a long follow-up, which allows for more robust outcome data.

In conclusion, OE is a safe therapy for controlling sinonasal and cutaneous malignancies invading the orbit when it is clinically indicated. Although OE is an aggressive approach, it is the authors' opinion that OE should be used judiciously in the setting of a multidisciplinary management plan based on tumor histology, the degree of orbital involvement, and consideration of alternative treatments. Finally, OE is a disfiguring procedure with negative social ramifications, and the appropriateness of its use must be discussed with each patient on a case-by-case basis in the context of the disease status.

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AUTHOR CONTRIBUTIONS

Jeffrey I. Traylor: Formal analysis, writing–original draft, and writing–review and editing. **Lana D. Christiano:** Data curation and writing–original draft. **Bita Esmaeli:** Writing–review and editing. **Matthew M. Hanasono:** Writing–review and editing. **Peirong Yu:** Writing–review and editing. **Dima Suki:** Formal analysis. **Wen Zhang:** Formal analysis. **Shaan M. Raza:** Writing–review and editing. **Ehab Y. Hanna:** Writing–review and editing. **Franco DeMonte:** Conceptualization, methodology, project administration, and writing–review and editing,

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